

Hotel Booking Analysis Dashboard Using Power BI

Objective

The objective of this project was to analyse hotel booking data using SQL, Power BI, and Python to identify booking trends, customer behaviour, cancellation patterns, and revenue-related insights. The dashboard was designed to support data-driven decision-making through interactive visualizations and key performance indicators (KPIs).

Dataset

The project used the **Hotel Booking Demand** dataset containing hotel reservation records from City Hotels and Resort Hotels.

The dataset included information such as:

- Hotel Type
- Booking Date
- Lead Time
- Customer Type
- Market Segment
- Distribution Channel
- Country
- Meal Type
- Room Type
- Average Daily Rate (ADR)
- Booking Status (Cancelled / Not Cancelled)

Tools Used

- SQL (MySQL)
- Power BI
- Power Query
- Python (Pandas)

Data Cleaning

The following data cleaning steps were performed before analysis:

- Removed duplicate records.
- Replaced missing values in relevant columns.
- Corrected data type issues.
- Replaced invalid values in the Children column.
- Verified date fields and numeric columns.
- Validated cleaned data using SQL, Power BI, and Python.

Business Questions Solved

The following business questions were analysed:

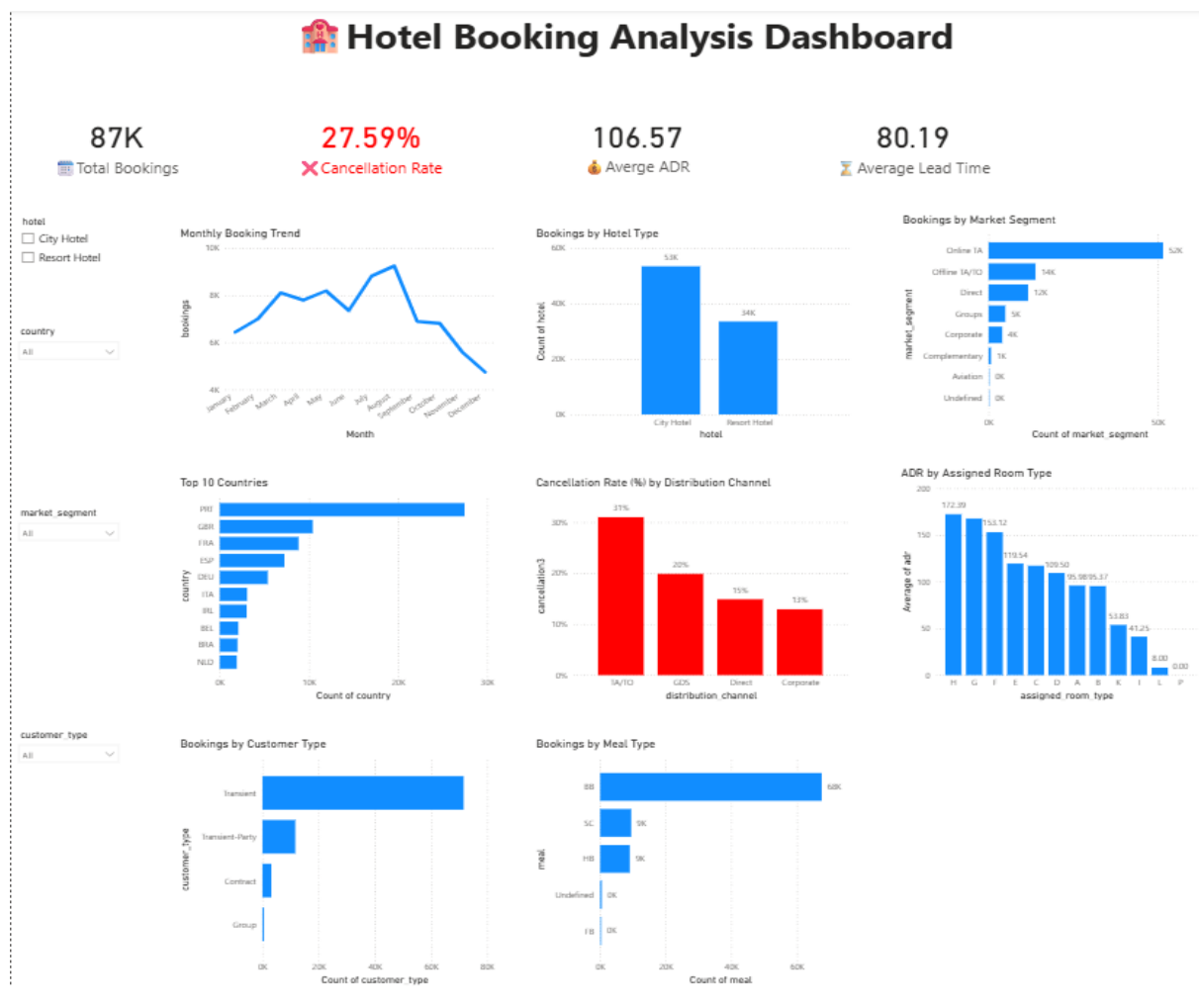
1. Which hotel has the highest number of bookings?
2. What is the overall cancellation rate?
3. Which month has the highest number of bookings?
4. Which country has the highest number of bookings?
5. Which hotel has the highest Average Daily Rate (ADR)?
6. Which hotel has the highest cancellation rate?
7. Which market segment has the highest number of bookings?
8. Which market segment has the highest cancellation rate?
9. Which customer type has the highest number of bookings?
10. Which assigned room type has the highest Average Daily Rate (ADR)?
11. Which distribution channel has the highest number of bookings?
12. Which distribution channel has the highest cancellation rate?
13. Which meal type is most preferred by customers?
14. Which meal type has the highest cancellation rate?
15. Which hotel has the highest average lead time?

Dashboard Components

The dashboard includes the following visualizations:

- KPI Cards
- Monthly Booking Trend
- Bookings by Hotel Type
- Bookings by Country
- Market Segment Analysis
- Customer Type Analysis
- Distribution Channel Analysis
- Average Daily Rate (ADR) Analysis
- Cancellation Rate Analysis
- Lead Time Analysis
- Interactive Slicers

Dashboard



Key Insights

- City Hotel recorded the highest number of bookings.
- Overall cancellation rate was approximately **27.59%**.
- August recorded the highest number of bookings.
- Portugal (PRT) generated the highest number of bookings.
- City Hotel recorded the highest Average Daily Rate (ADR).
- City Hotel had a higher cancellation rate than Resort Hotel.
- Online TA generated the highest number of bookings among all market segments.
- Online TA also recorded the highest cancellation rate among major market segments.
- Transient customers accounted for the highest number of bookings.
- Assigned Room Type H recorded the highest Average Daily Rate.
- TA/TO was the most frequently used distribution channel.
- TA/TO recorded the highest cancellation rate among major distribution channels.
- BB was the most preferred meal type.
- SC meal type recorded the highest cancellation rate.
- Resort Hotel customers booked significantly earlier, recording the highest average lead time.

Conclusion

This project demonstrates how SQL, Power BI, and Python can be integrated to perform complete data analysis. SQL was used to answer business questions, Python was used to validate analytical results, and Power BI was used to build interactive dashboards and visualizations. The project provided meaningful insights into hotel booking behaviour, customer preferences, cancellation trends, pricing, and booking patterns, enabling data-driven decision-making.